

David Vayntrub

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University of California, Riverside

Computer Science, Expected Graduation: 2027

Experiences:

- **StudySpot – Cofounding Engineer** **May 2025 - Sep 2025**
 - AI-based study tool that uses school documents as context with the **Anthropic LLM** built using a **Next.js** and **Typescript** frontend and **Supabase** backend
 - Integrated **Canvas LMS API** to determine courses and upload documents based on API key, decreasing new user onboarding time by 50 %
 - Improved document ingestion speed by 30% by integrating **LlamaParse**
- **ACM Riverside - Forge – Software Engineer** **Jan 2025 - Mar 2025**
 - Collaborated on a 10-member team to develop a self-moving chess set
 - Utilized **Python SpeechRecognizer** and **PyAudio** to detect and parse voice command
 - Compared **whisper C++** speech-to-text model against **Python SpeechRecognizer** for speech accuracy to determine which to implement
- **Citrus Hack 2025 – Frontend Developer** **Apr 2025**
 - Created an AI-based movie recommendation platform in a 36-hour hackathon
 - Used **Next.js** and **Typescript** for frontend with **Convex** backend/**Clerk** for authentication
 - Implemented cosine similarity algorithm in **Python** to generate user-specific suggestions
- **Cutie Hack 2025 – Frontend Developer** **Nov 2025**
 - Built a **Next.js** educational platform in a 12-hour hackathon to teach CS fundamentals to K-12 students using the **Blockly Library**.
 - Coded 15 custom interactive logic blocks backed by real-time JavaScript execution

Projects:

- **Analysis of Machine Learning Methods with Regression** **Aug 2022 - Mar 2024**
 - Analyzed classification models using **Jupyter Notebook**: Decision Trees, Logistic Regression, SVM, KNN
 - Used **Python's numpy** and **pandas** libraries to visualize and handle data
 - Wrote 20-page paper analyzing the effectiveness of the algorithms based on multivariable datasets influencing income
- **Inventory Management System** **Jun 2025 - Jul 2025**
 - Individually designed a RESTful inventory management API using **Java Spring Boot**
 - Implemented CRUD functionality, category-based filtering, and integration with **MySQL** for persistence.
- **Task Terry** **Apr 2025 - Jun 2025**
 - Developed a full-stack schedule management web app using **C++** backend integrated with **Node.js/Express** and a **React + Tailwind CSS** frontend
 - Applied Agile and SOLID principles through Kanban-based sprint planning
 - Implemented unit tests for backend components using **Google Test**, ensuring high reliability across **C++–JavaScript** communication via **Express** middleware

Skills:

Languages & Tools: Python, TypeScript, JavaScript, Java, C++, SQL, Tailwind CSS, Jupyter, Git, Figma

Frameworks & APIs: Next.js, Supabase, Convex, Clerk, SpeechRecognizer, Whisper, Spring Boot

Other: Data Analysis, AI Integration, UI/UX Design, Realtime Systems, Voice Recognition, MySQL